



BITS Pilani

Hyderabad Campus

Analog Electronics

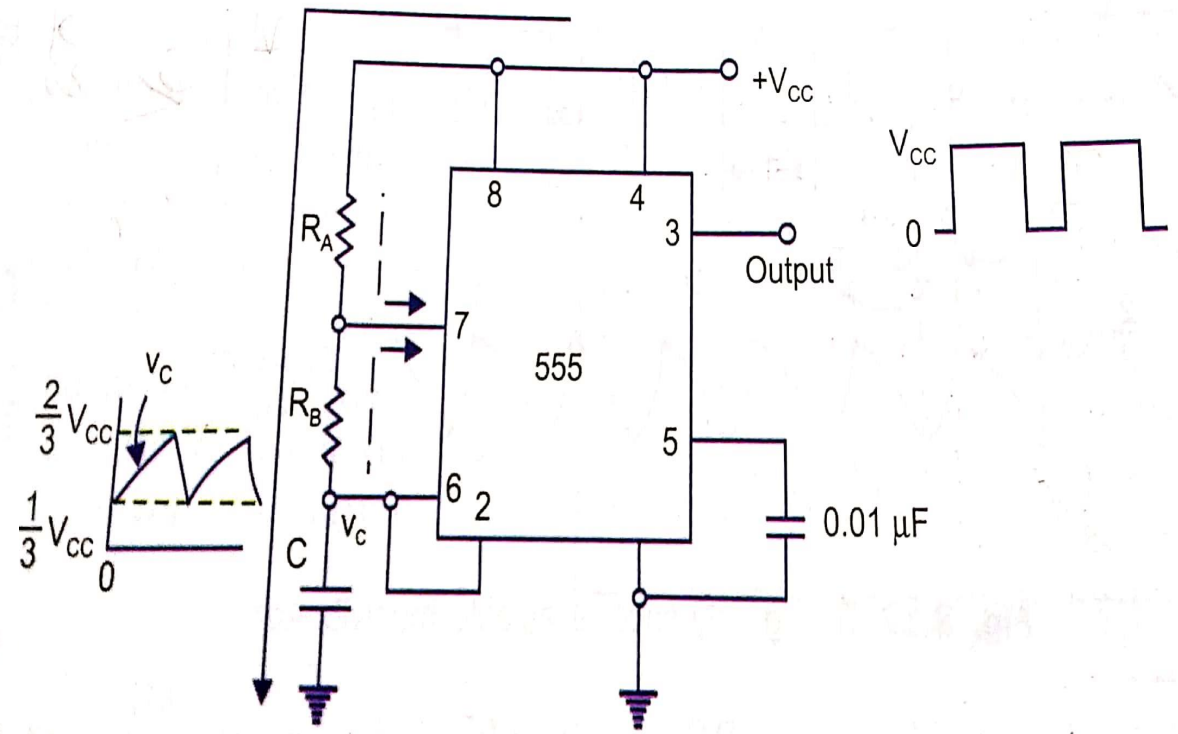
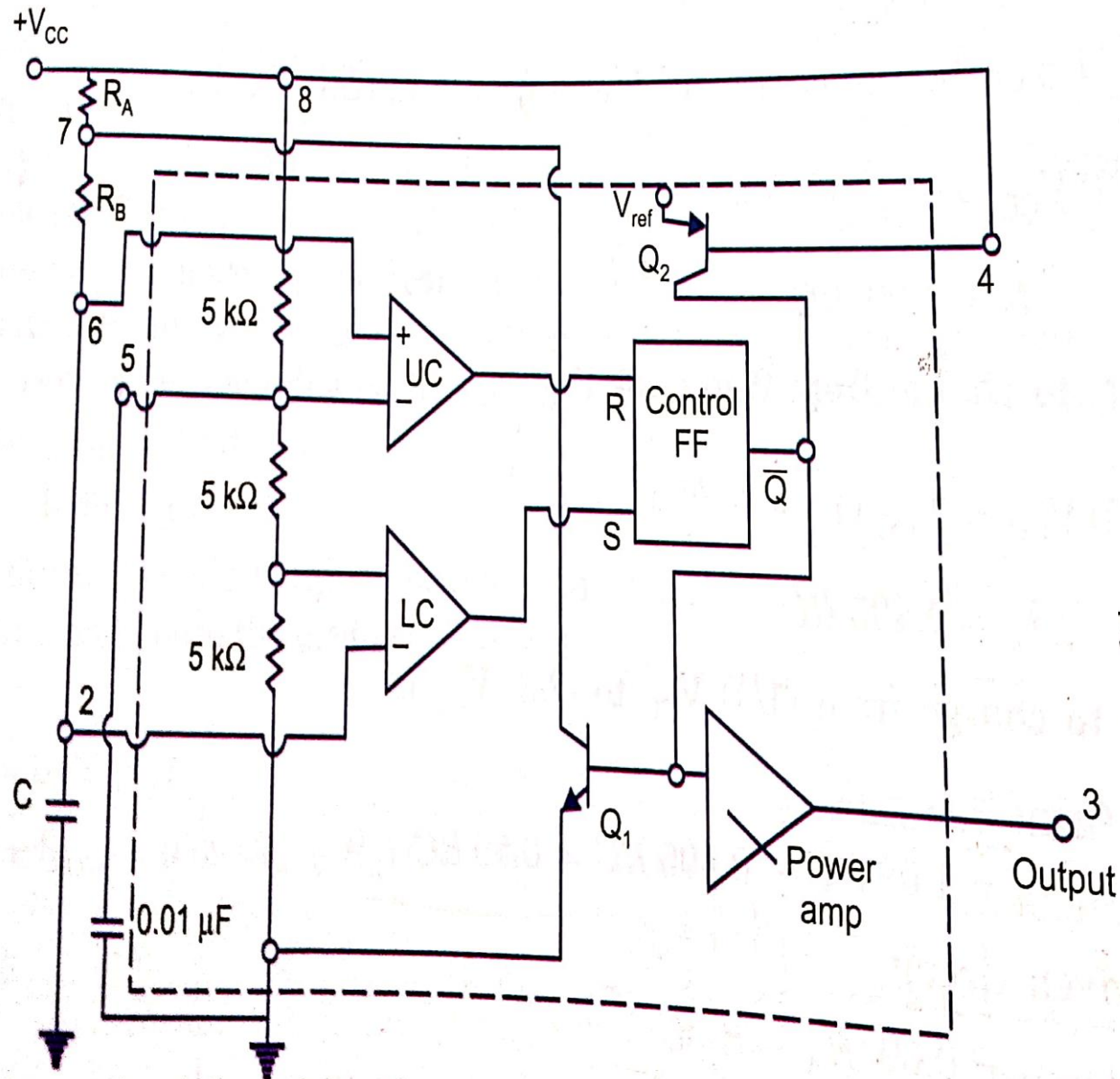
Lecture-33: IC 555 Timer (Square Wave Generator, VCO)

Prof. Sk Masum Nawaz

nawaz.masum@hyderabad.bits-pilani.ac.in



Astable operation of IC 555



Astable operation of IC 555

① Initially, the cap C was uncharged ($V_C = 0$), Pin-2 = 0. Comp-2 output is high. IC output is high ($Q = 1$) $\overline{Q} = 0$. Transistor Q_1 is off.

② The timing capacitor will start charging toward V_{CC} through R_A and R_B . Charging time constant is $(R_A + R_B)C$.

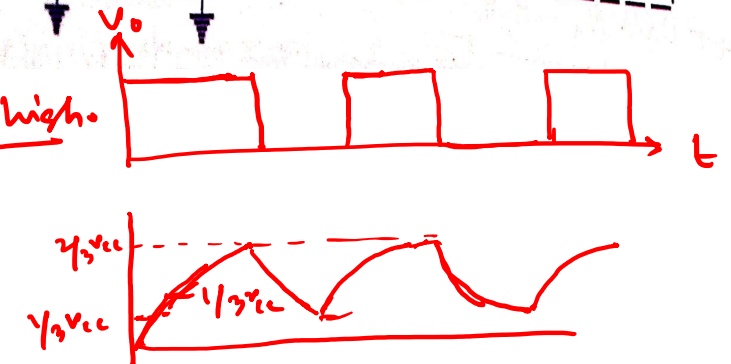
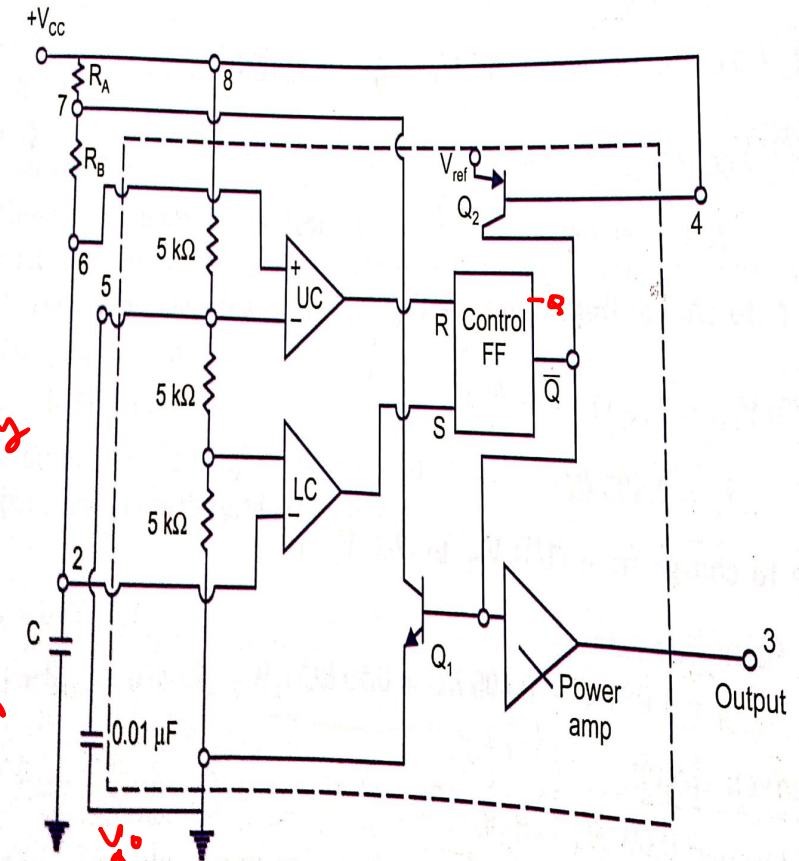
③ When $V_C > \frac{1}{3}V_{CC}$, Comp-2 goes to high ($S = 1$). This will not affect the IC output (pin-3) as it is already at high state.

④ When $V_C > \frac{2}{3}V_{CC}$, comp-1 goes high ($R = 1$) and output will go to low. $Q = 0$, $\overline{Q} = 1$. Transistor Q_1 is ON. $\therefore$ Cap C will discharge from $\frac{2}{3}V_{CC}$ to zero through R_B and Q_1 . Discharge time constant $\propto R_B \cdot C$

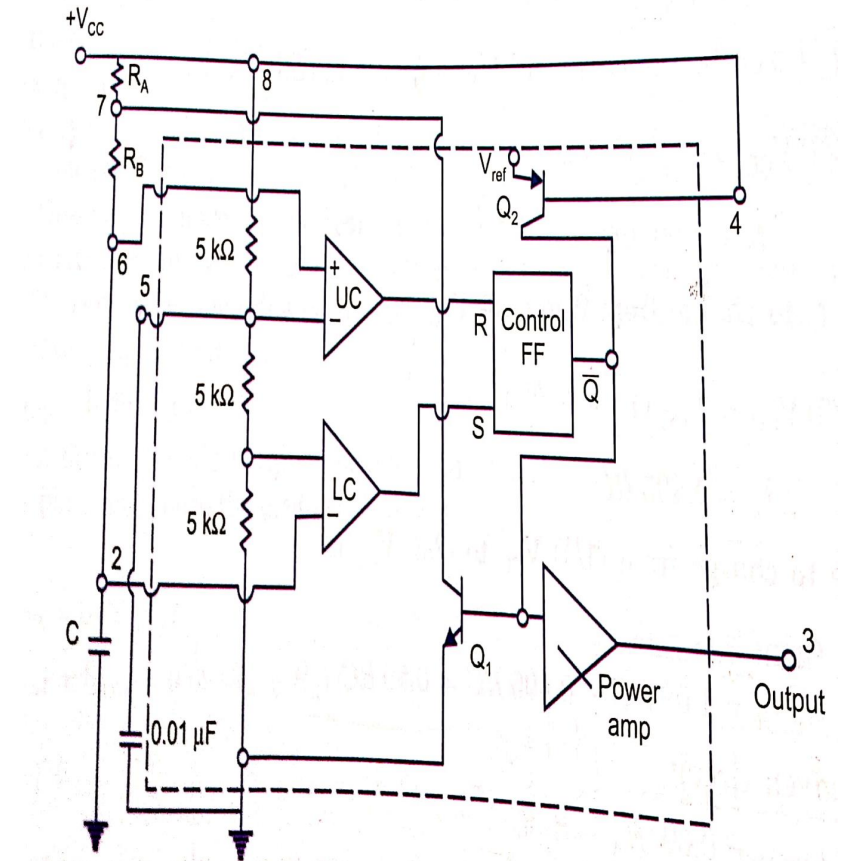
⑤ During this discharge when $V_C < \frac{1}{3}V_{CC}$, comp-2 will go to high. ($S = 1$). Output of the IC (pin-3) will go to high.

$\therefore Q = 1$, $\overline{Q} = 0$. Transistor Q_1 is off. Cap C starts charging again toward V_{CC} through R_A and R_B

⑥ And the cycle repeats.



Astable operation of IC 555

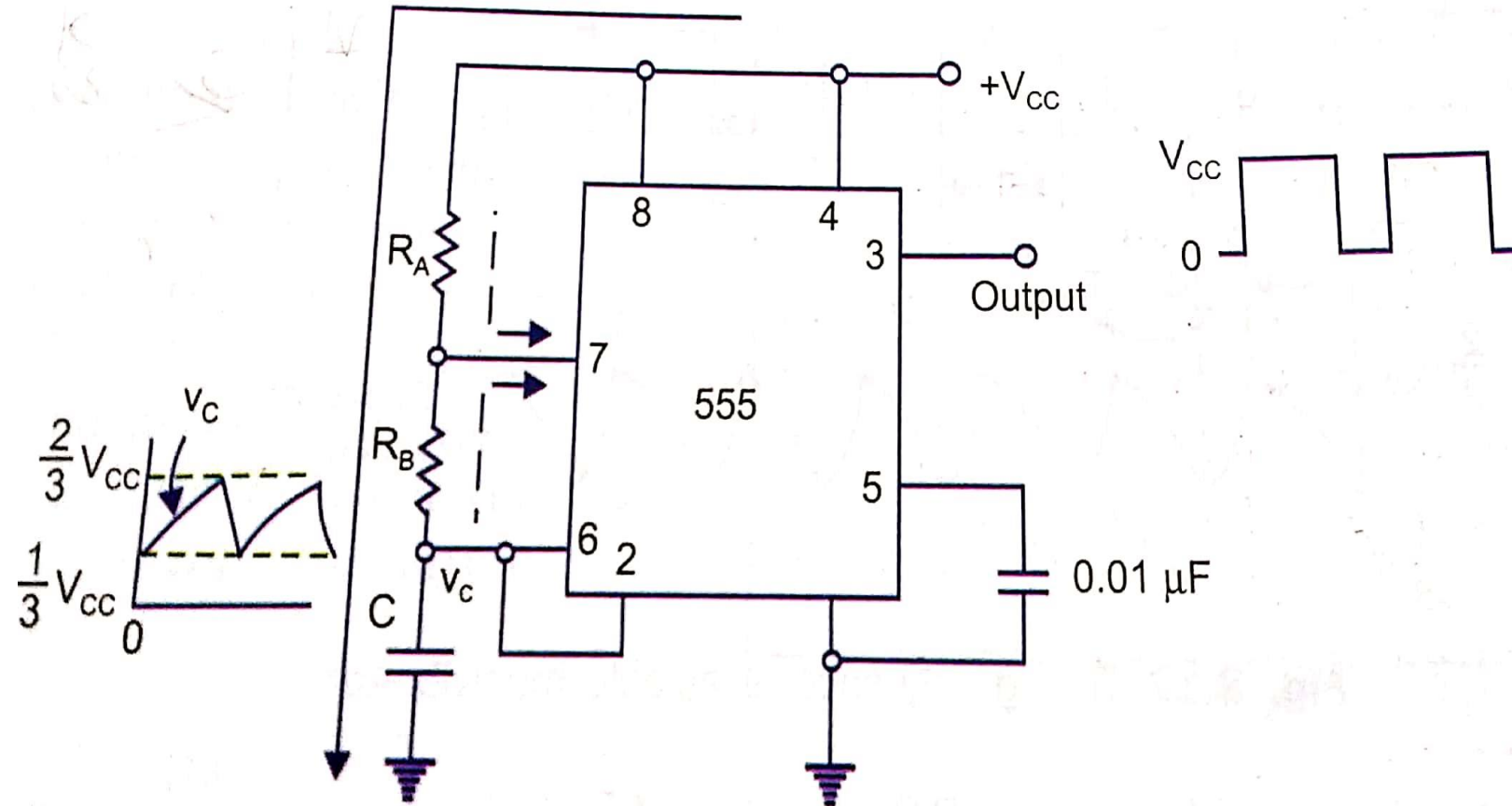


Astable operation of IC 555

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Astable operation of IC 555—Calculation of Time period, Duty cycle

On time - (T_H)

$$V_c(t) = V_{\infty} - (V_{\infty} - V_{0+}) e^{-t/\tau_H}, \quad \tau_H = (R_A + R_B)C$$

$$V_{\infty} = V_{CC}, \quad V_{0+} = \frac{1}{3}V_{CC}$$

$$\text{At } t = T_H, \quad V_c(T_H) = \frac{2}{3}V_{CC}$$

$$\frac{2}{3}V_{CC} = V_{CC} - (V_{CC} - \frac{1}{3}V_{CC}) e^{-T_H/\tau_H}$$

$$\Rightarrow T_H = \tau_H \ln(2) \\ = 0.693(R_A + R_B)C$$

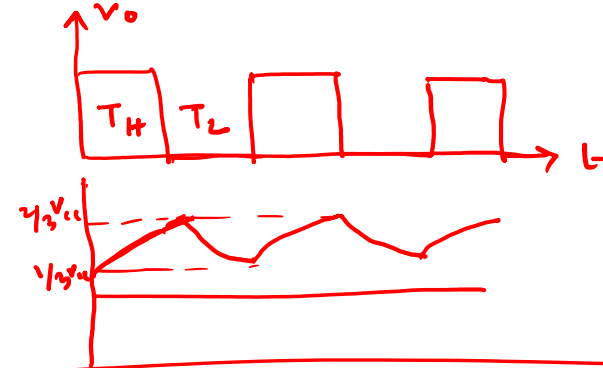
Off time (T_L)

$$V_c(t) = V_{\infty} - (V_{\infty} - V_{0+}) e^{-t/\tau_L}, \quad \tau_L = R_B \cdot C$$

$$V_{\infty} = 0V, \quad V_{0+} = \frac{2}{3}V_{CC} \quad \text{At } t = T_L, \quad V_c(T_L) = \frac{1}{3}V_{CC}$$

$$\frac{1}{3}V_{CC} = 0 - (0 - \frac{2}{3}V_{CC}) e^{-T_L/\tau_L}$$

$$T_L = \tau_L \ln(2) = 0.693 R_B \cdot C$$



$$\text{Time period } T = T_L + T_H \\ = 0.693(R_A + 2R_B)C$$

$$\text{Frequency } f = \frac{1}{T}$$

$$\text{Duty cycle (D)} = \frac{T_H}{T} \times 100\% \\ = \frac{R_A + R_B}{R_A + 2R_B} \times 100\%$$

$$R_A \rightarrow 0, \quad D \rightarrow 50\%$$

$$R_B \rightarrow 0, \quad D \rightarrow 100\%$$

$\therefore$ Duty cycle varies between $50 \rightarrow 100\%$

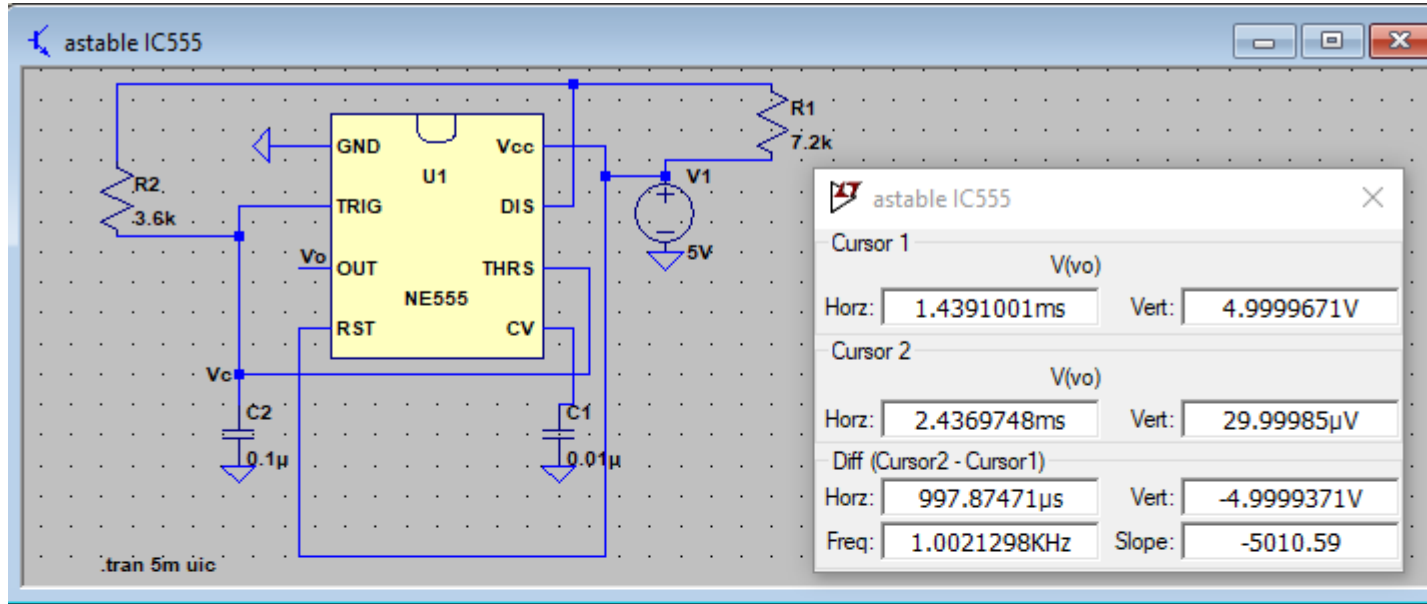
Can we achieve duty cycle exactly 50% ?

Astable operation of IC 555

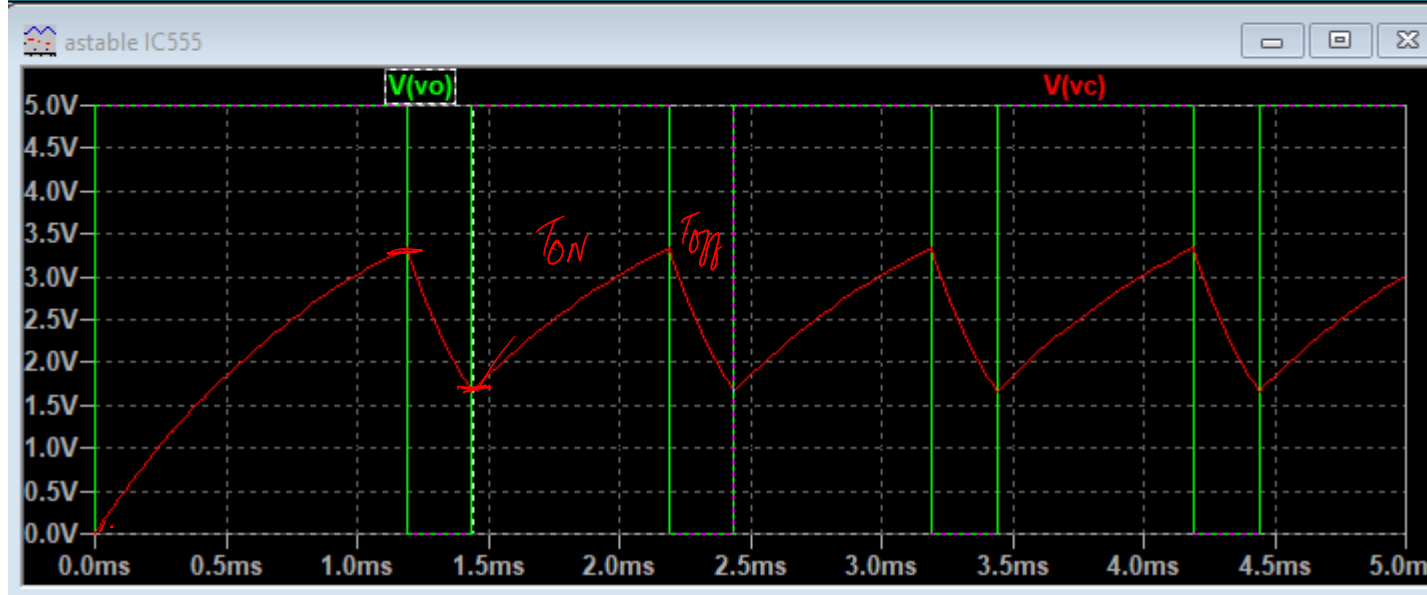
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$$(R_A + R_B) C$$
$$R_B C$$

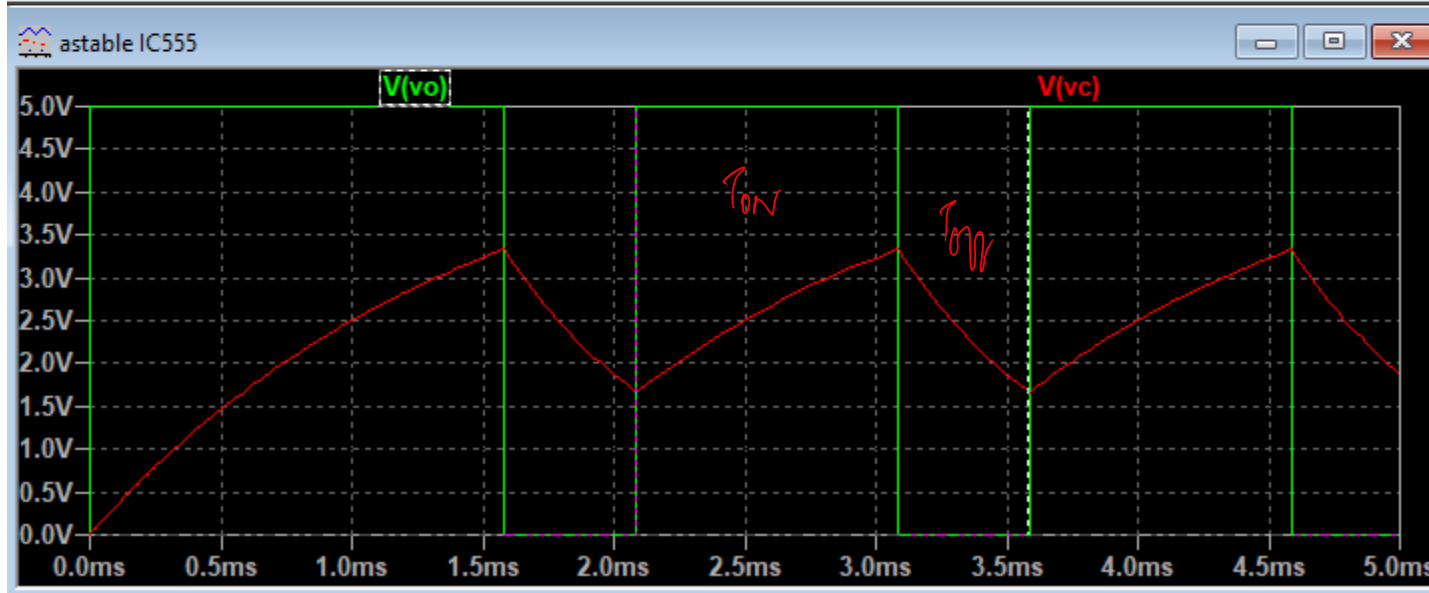
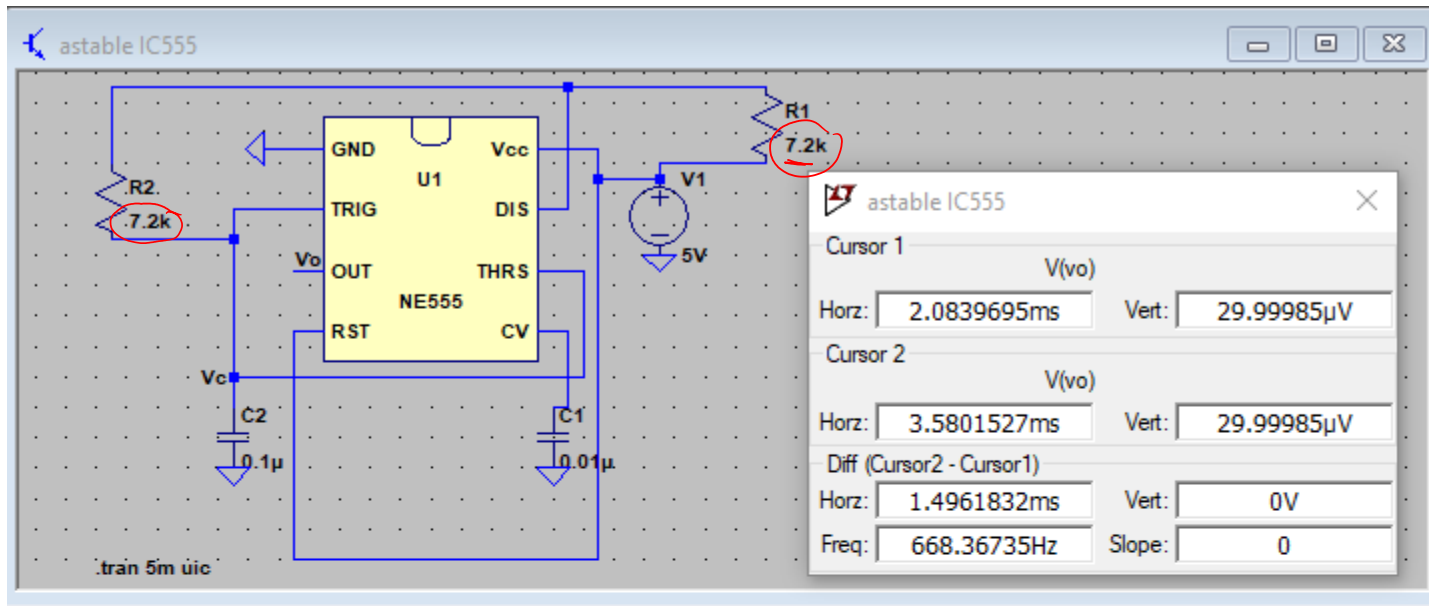


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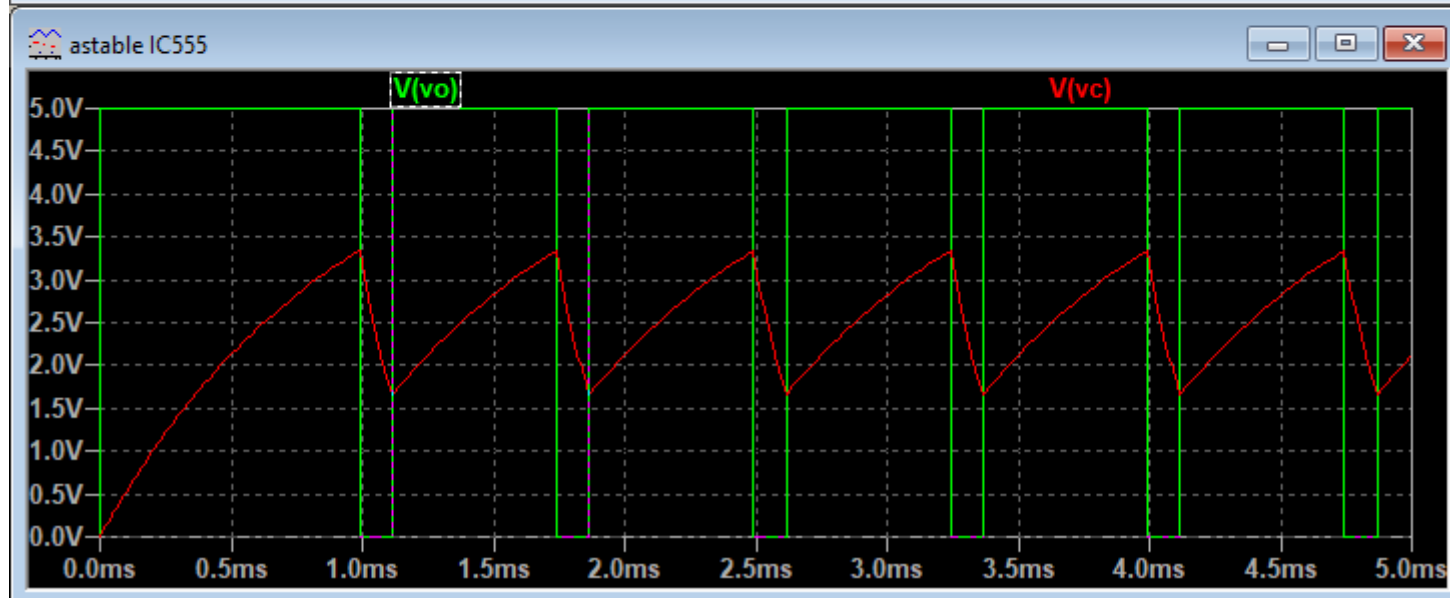
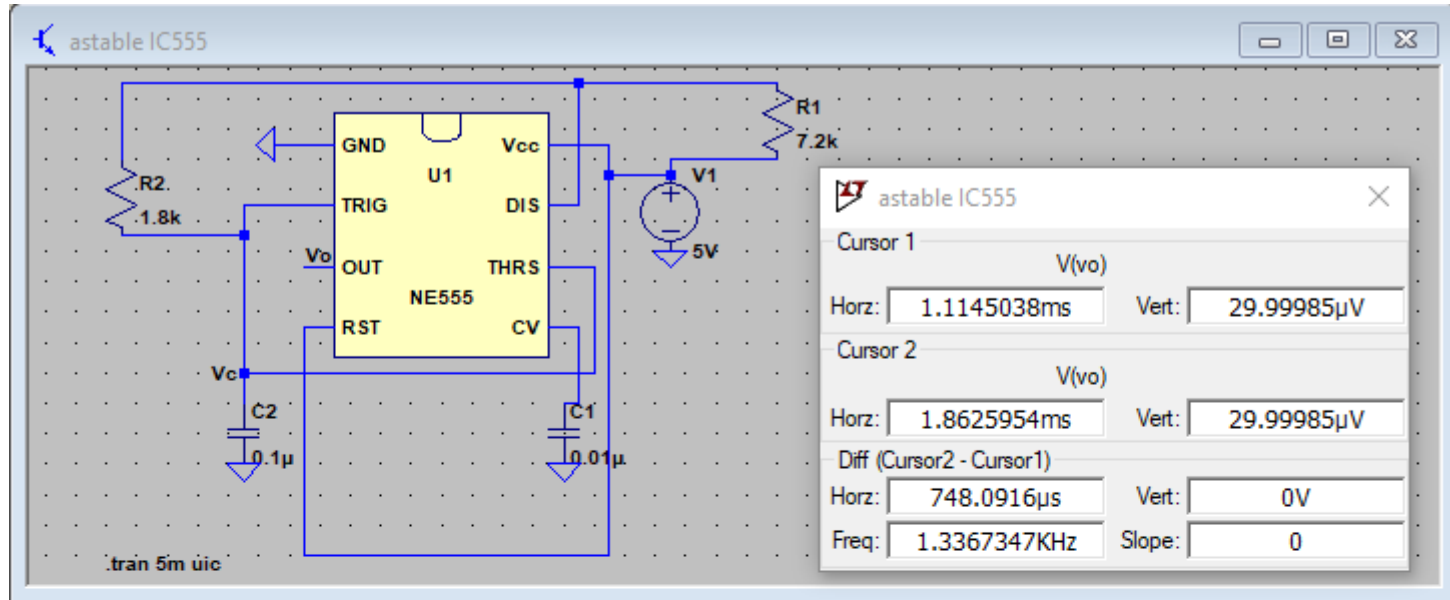


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Astable operation of IC 555



- As the timing capacitor, C charges through R_A and R_B ($T_H = 0.69 (R_A + R_B) C$) but only discharges through R_B ($T_L = 0.69 R_B C$) the output duty cycle $[D = (R_A + R_B) / (R_A + 2R_B)]$ can be varied between 50 to 100 % by changing the value of resistor R_B .
- By decreasing the value of R_B the duty cycle increases towards 100%.
- By increasing R_B the duty cycle reduces towards 50 %.
- **Hence the problem with this basic astable 555 oscillator configuration is that the duty cycle, will never go below 50%.**
- In order to obtain a square wave R_A should be reduced to zero which will connect pin 7 directly to V_{cc} and will damage transistor Q1.
- One way to overcome this problem is to connect a signal bypassing diode in parallel with resistor R_B .

Astable operation of IC 555 – Square wave generation

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To have ^{exact} 50% duty cycle

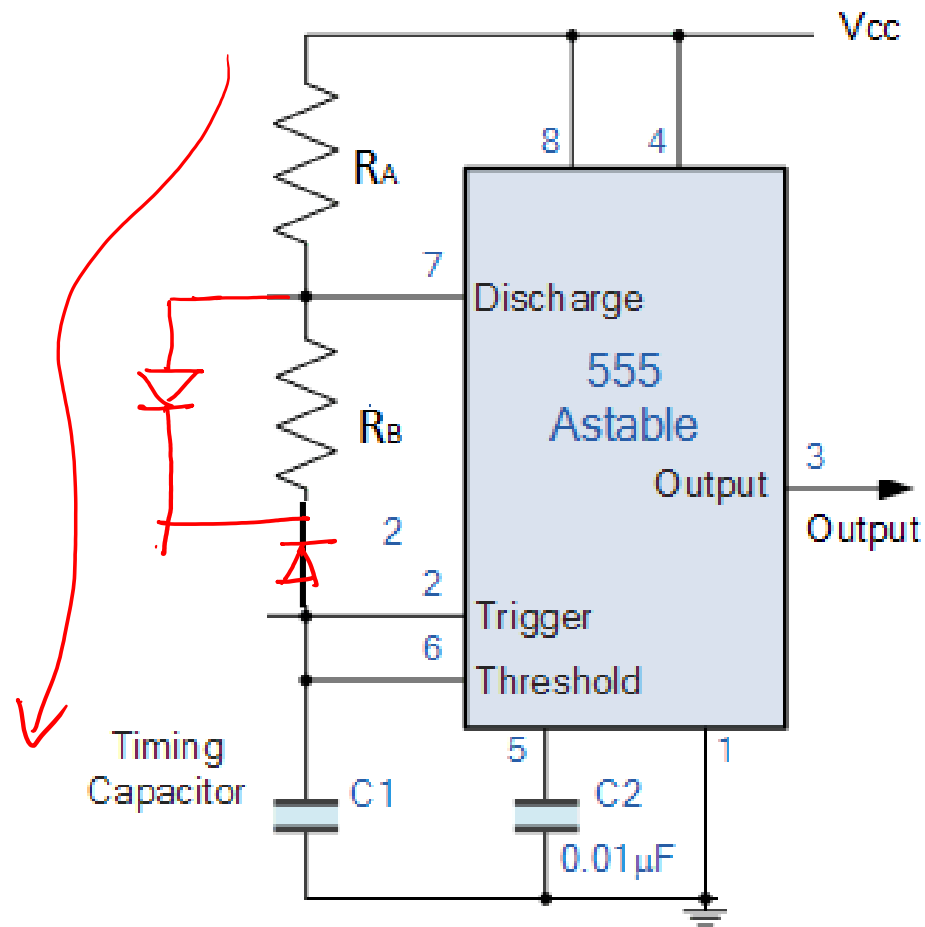
→ On time = OFF time

→ Charging time const
= Discharging time const.

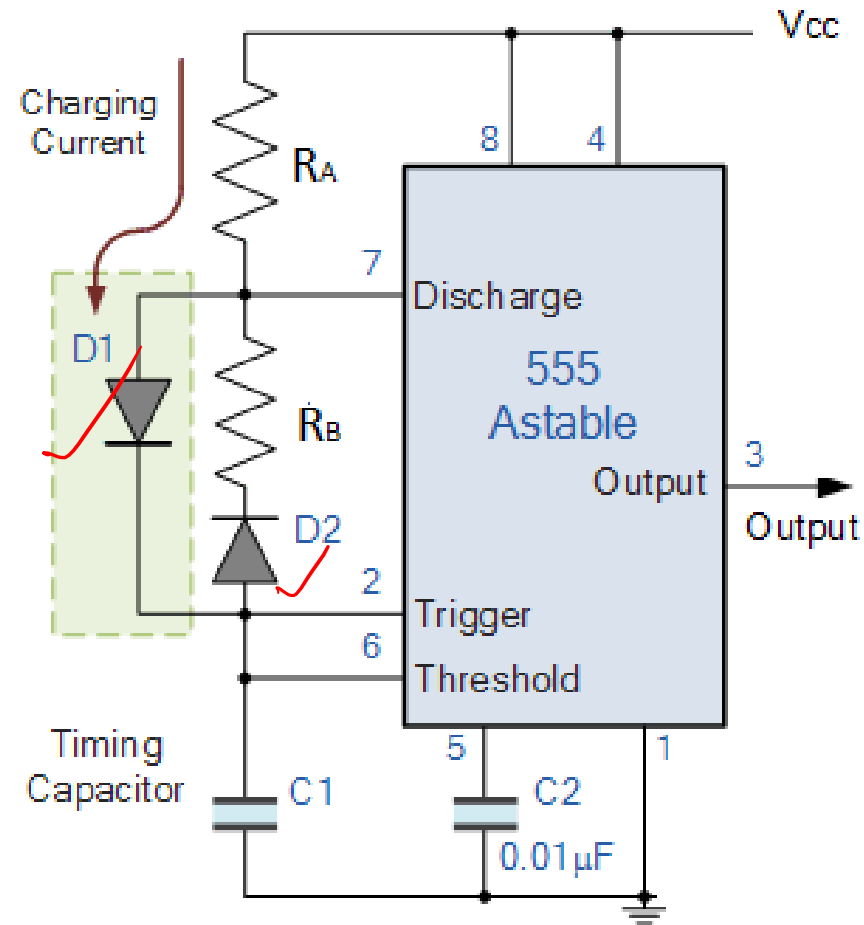
$$R_A = R_B$$

Charging time = Discharging time

Duty cycle = 50%



Astable operation of IC 555 – Square wave generation



$$R_A = R_B$$

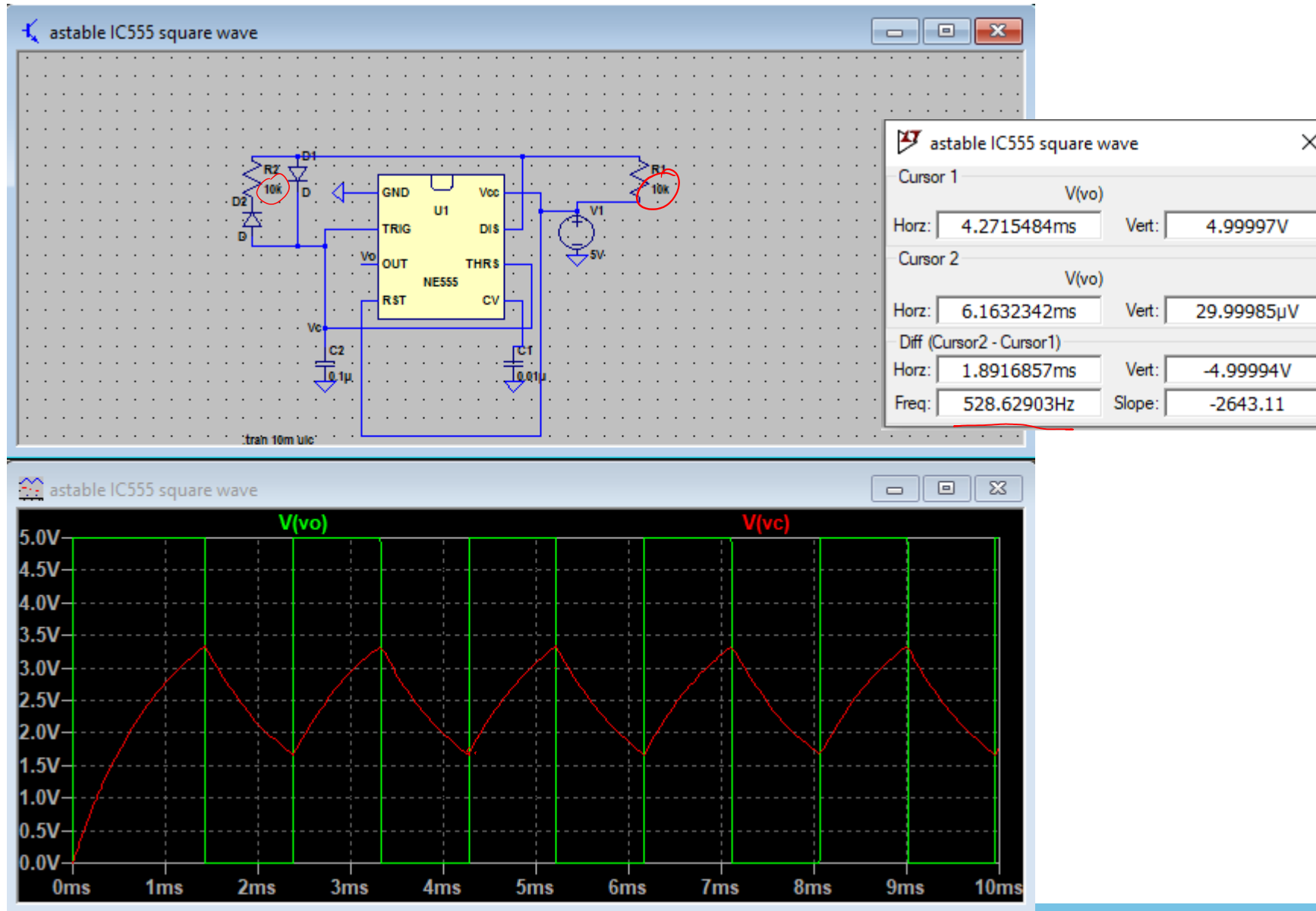
- Diode, D_1 connected between the trigger input and the discharge input.
- The timing capacitor will now charge through resistor R_A only, (as resistor R_B is effectively shorted out by the diode).
- The capacitor discharges as normal through resistor, R_B .
- An additional diode, D_2 can be connected in series with the discharge resistor, R_B is required to ensure that the timing capacitor will only charge through R_A and D_1 and not through the parallel path of R_B . (during the charging process diode D_2 is connected in reverse bias blocking the flow of current through itself).

Astable operation of IC 555 – square wave generation

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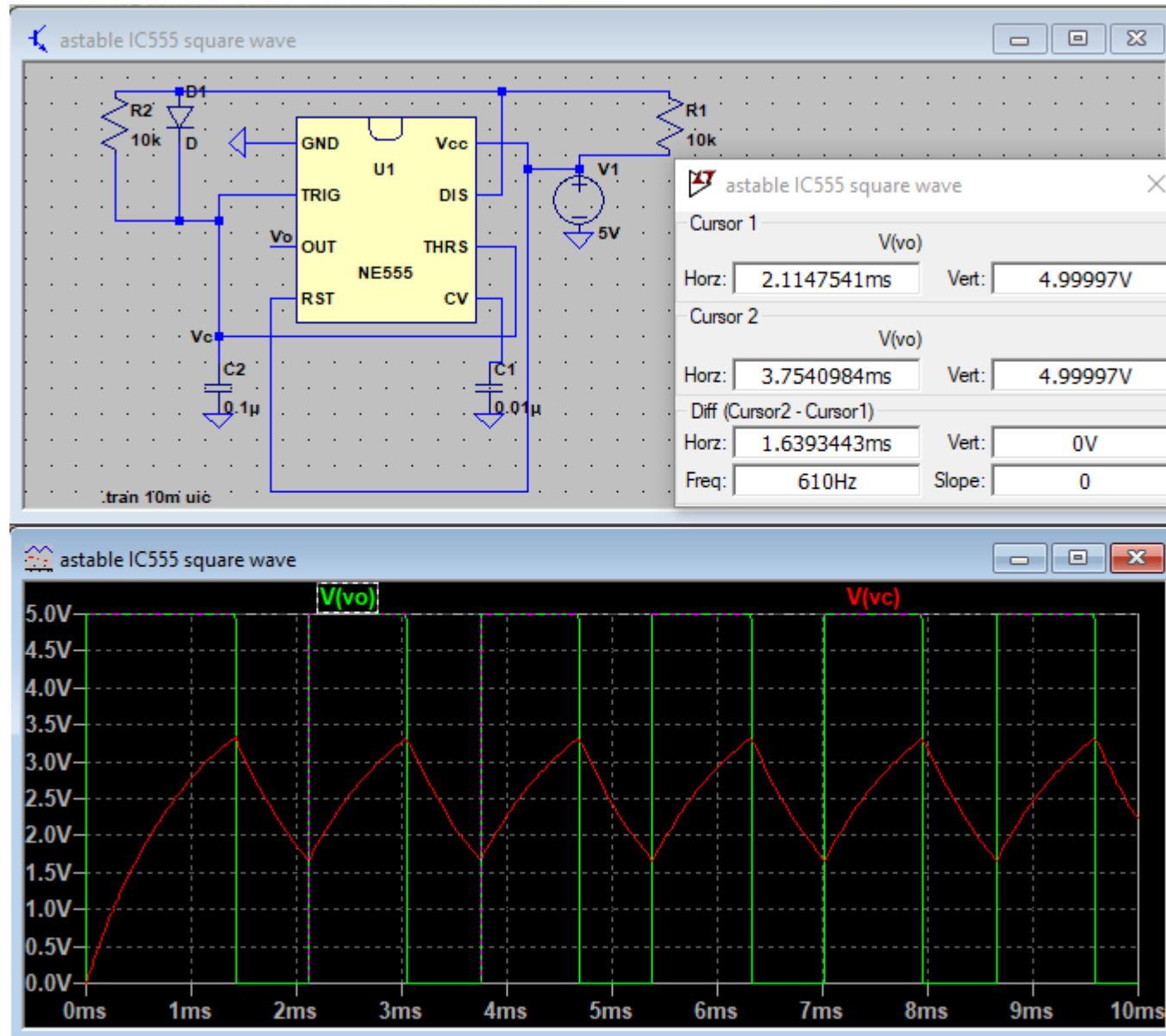


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Without diode D₂
duty cycle will not be
equal to 50 %

Find the values of R_A and R_B that result in oscillation of 100 kHz and duty cycle of 25 %. The timing capacitor $C=1000$ pF.

(consider the ext with diodes D_1 and D_2)
Neglect the diode ON resistance

$$T_{ON} = 0.693 R_A \cdot C$$

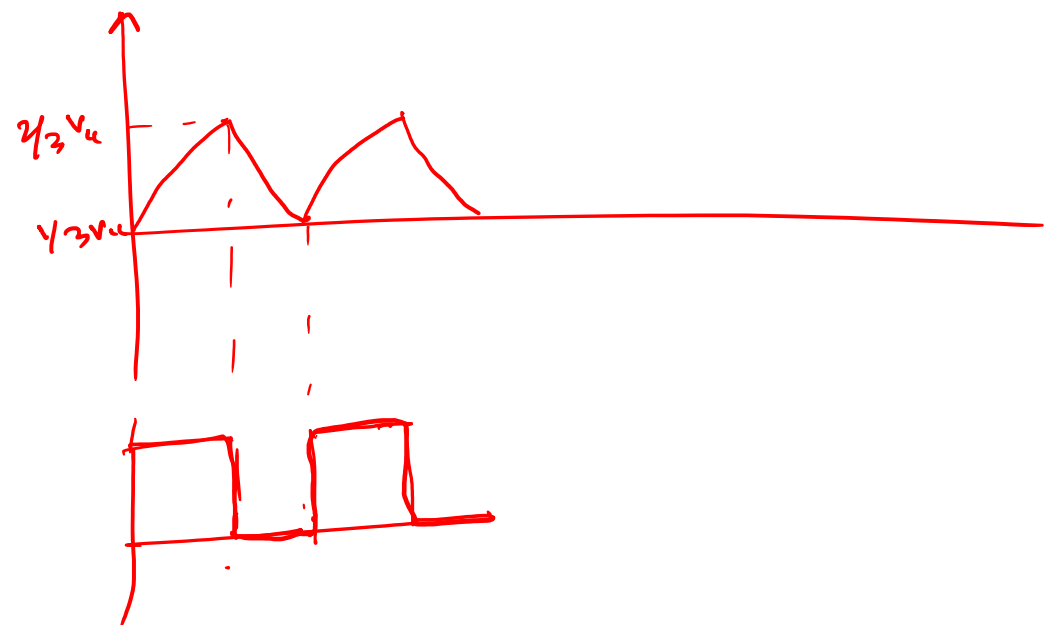
$$T_{OFF} = 0.693 R_B \cdot C$$

$$T = 0.693 (R_A + R_B) \cdot C = 10^{-5} \Rightarrow (R_A + R_B) = \frac{10^{-5}}{0.693 \times 10^{-9}}$$

$$\text{Duty cycle } D = \frac{T_{ON}}{T} = \frac{R_A}{R_A + R_B} = 0.25$$

$$R_A = 3.6 \text{ k}\Omega$$

$$R_B = 3 R_A = 10.8 \text{ k}\Omega$$

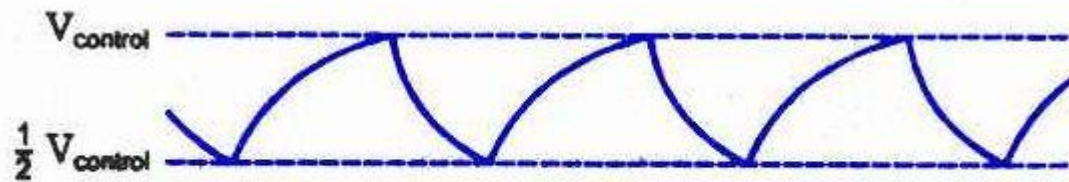
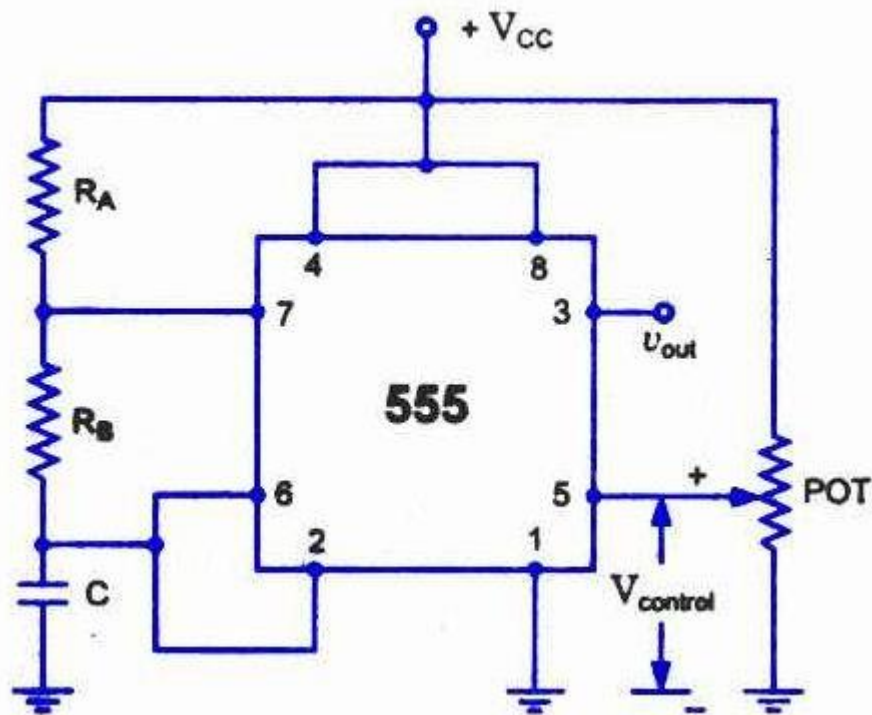


Voltage controlled oscillator, VCO



Output frequency will be a function of input/control voltage $\Rightarrow$ VCO

- A **Voltage Controlled Oscillator** is an oscillator which produces oscillating signals (waveforms) with variable frequency.
- The frequency of this waveform is varied by varying the magnitude of the Input voltage.
- Voltage Controlled Oscillator (VCO) to be a black box which takes in Voltage of variable magnitude and produces an output signal of variable frequency, and the frequency of the output signal is proportional to the magnitude of the input voltage.
- **Types of VCO circuits**
 - A very basic one can be built by just utilizing a capacitor, inductor and resistor to make a tank circuit.
 - Also Op-Amps, Multivibrator, transistors, 555 timers can be utilized to build **VCO**.
 - Dedicated IC packages like LM566 LM567 etc. can also act as VCO.

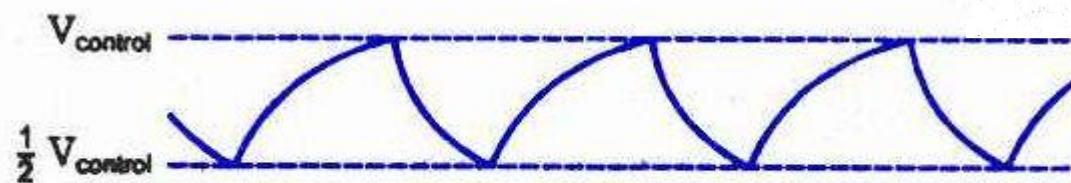
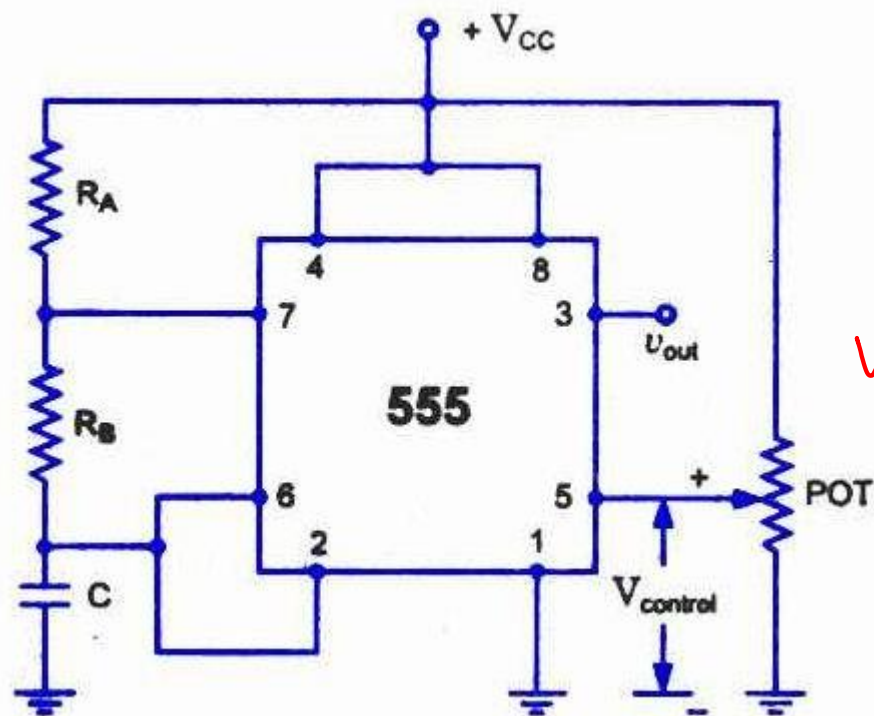


Timing Capacitor Voltage Waveform

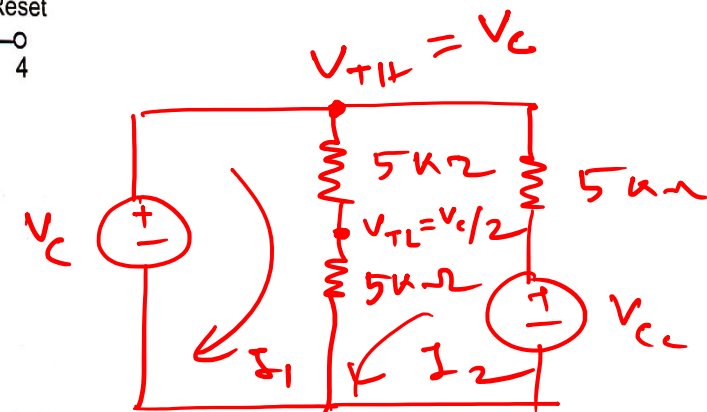
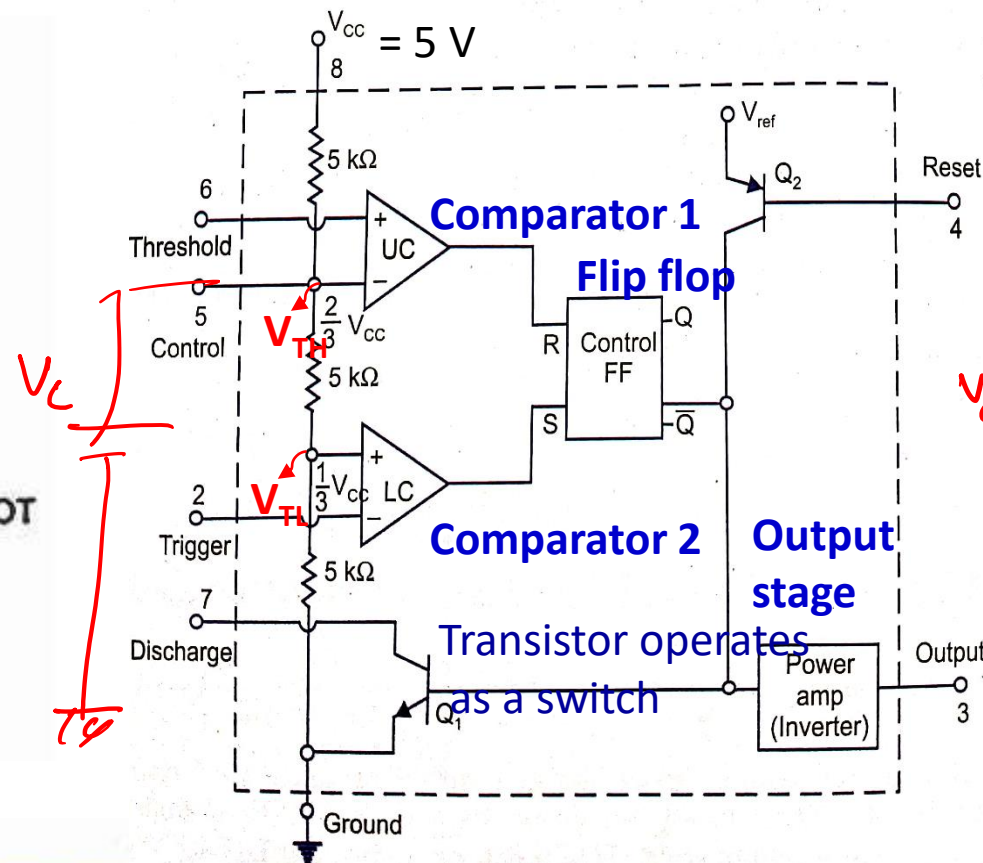
VCO

- * Pin No. 5 is important
- * It is the control pin
- * Normally it will be connected to ground but in case of operating IC 555 timer as VCO pin. 5 should be connected to external voltage
- * Either directly to external voltage or by connecting to V_{cc} itself using POT so that control voltage can be varied.

~~✗~~



Timing Capacitor Voltage Waveform

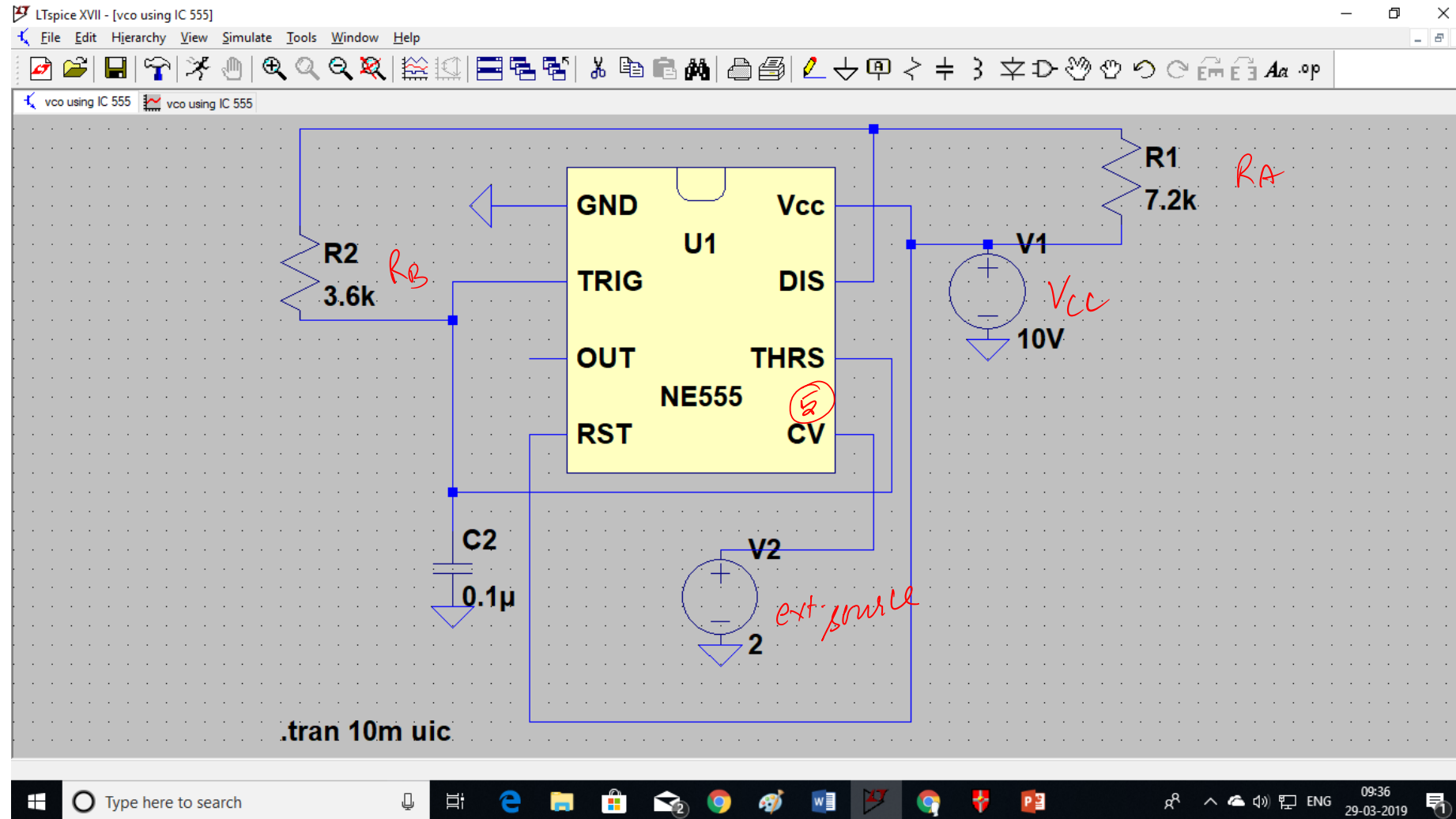


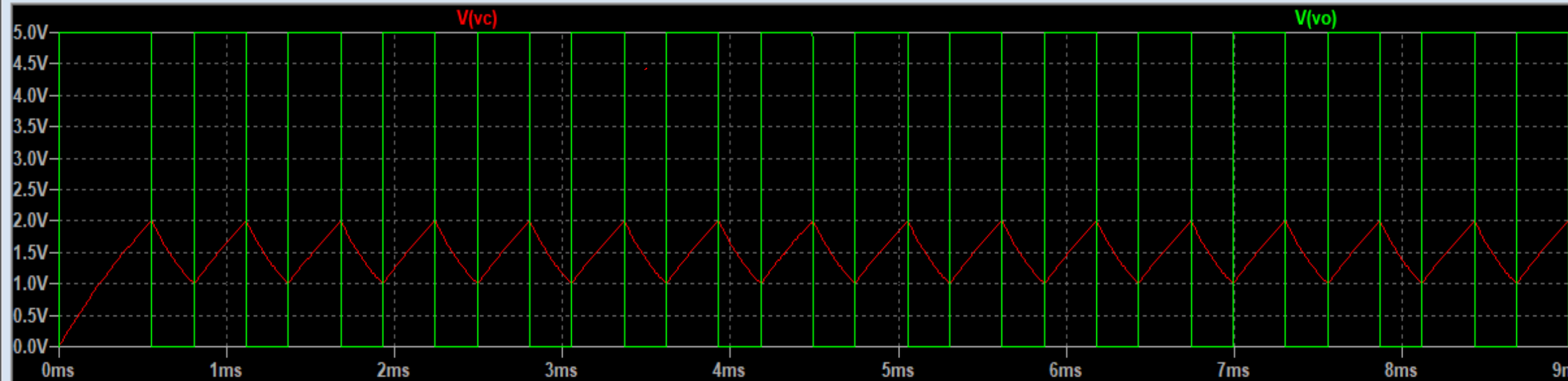
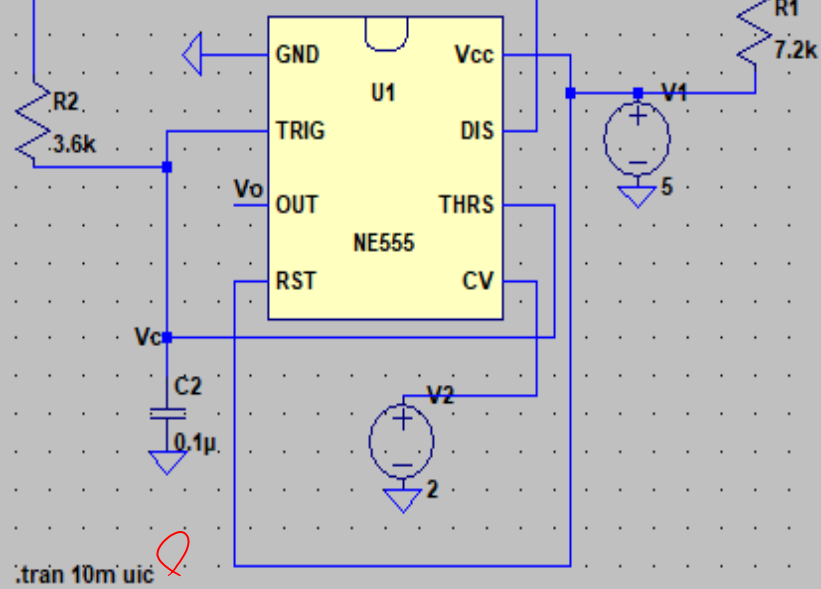
Now, the threshold voltages V_{TH} and V_{TL} are independent of V_{cc} and it completely dependent on V_c

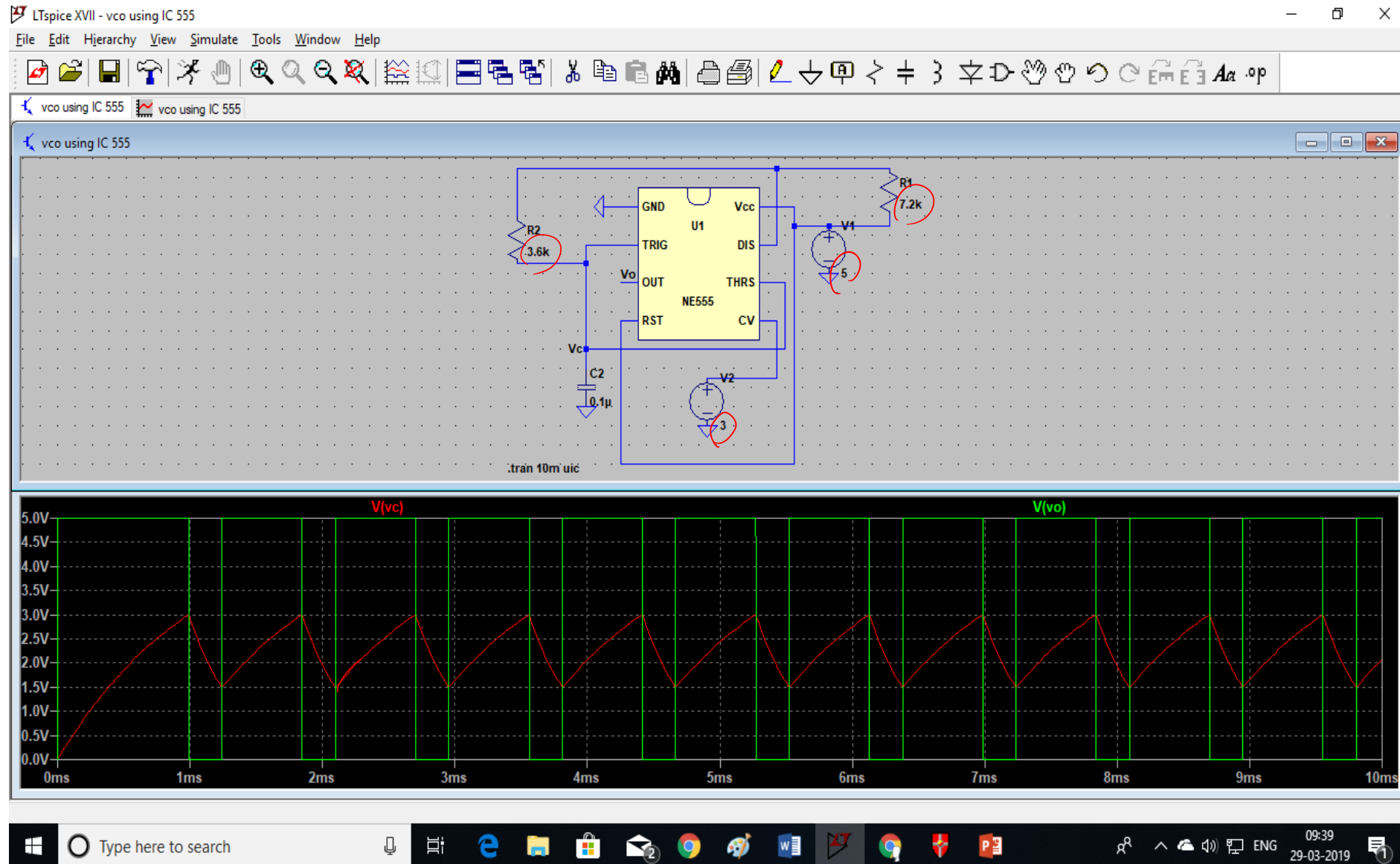
$$V_{TH} = V_c$$

$$V_{TL} = V_c/2$$

- VCO is voltage-to-frequency converter because the output frequency can be changed by changing the input voltage.
- pin 5 in IC 555 is control voltage terminal and its function is to control the threshold and trigger levels.
- Normally, the ~~control~~ ^{V_{TH}} voltage is $+2/3V_{CC}$ because of the internal voltage divider.
- However, an external voltage can be applied to pin 5 directly or through a pot, and by adjusting the pot, control voltage can be varied.
- Voltage across the timing capacitor varies between $+V_{control}$ and $1/2 V_{control}$ instead of $2/3 V_{CC}$ and $1/3 V_{CC}$.
- If control voltage is increased, the capacitor takes a longer time to charge and discharge; the frequency, therefore, decreases.
- Thus the frequency can be changed by changing the control voltage.

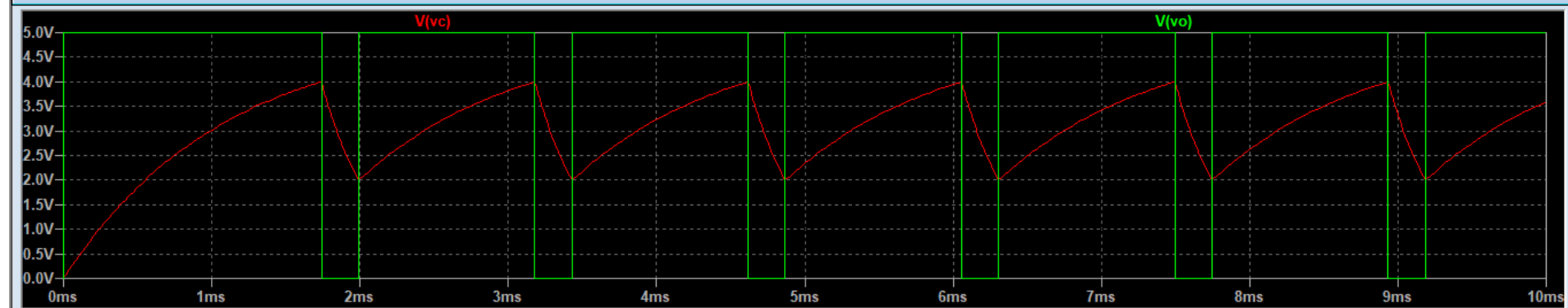
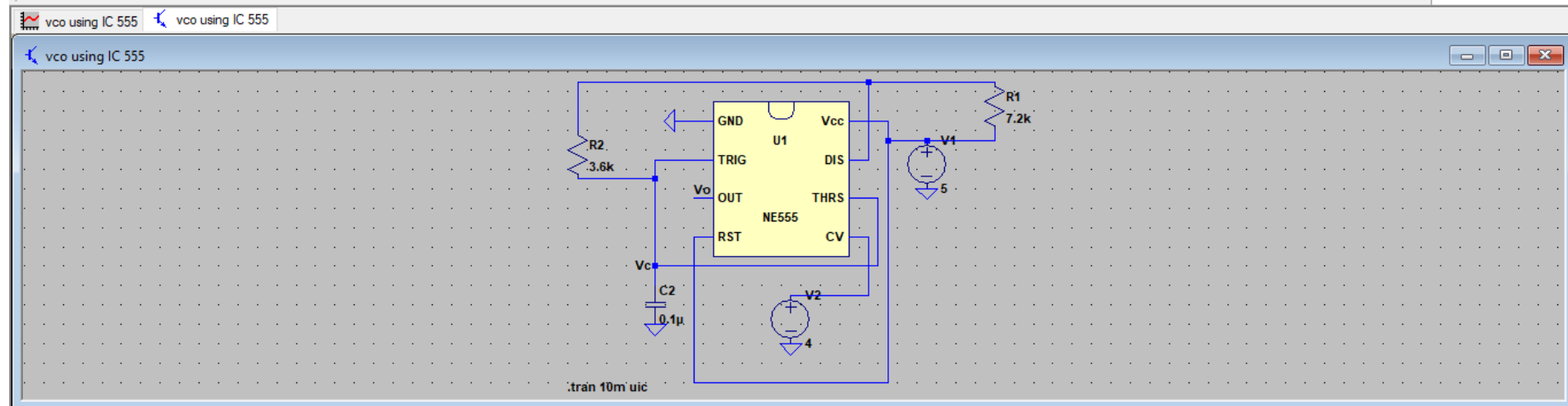






LTspice XVII - vco using IC 555

File Edit Hierarchy View Simulate Tools Window Help



x = 10.00ms y = -0.070V

1 new notification



- Frequency Shift Keying
- Frequency identifiers
- Keypad Tone recognizers
- Clock/Signal/Function Generators
- Used to build Phase Locked Loops.

